UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SAIA 1013: Responsible AI and Ethics

SECTION 01
Introduction to Responsible AI and AI Ethics

Course Lecturer: Dr. Nurazean Maarop
Whatsapp:0169853093
Email: nurazean.kl@utm.my
Room: MJIIT 060601 (Level 6)

COURSE OVERVIEW



Course Synopsis



Course Learning Outcomes



Course Evaluation



Course Contents

This course addresses the ethical considerations and responsibilities that must be taken into account concerning artificial intelligence. With appropriate ethics and responsibility, the objective of this course is to equip students with the knowledge and guided skill set necessary to develop AI for the benefit of society and technology consumers. In addition, the course underlines the significance of comprehending the risk, impacts and threats posed by new developments in AI technology, as well as the application of present regulations. Ethical issues in AI, AI governance, AI system transformation, ethical risk assessment and management, ethical matrices, along with societal and educational AI ethics will be among the subjects covered.

COURSE OVERVIEW



Course Objectives



Course Learning Outcomes



Course Evaluation



Course Contents

No.	CLO*	PLO (MQF Cluster Code)	Taxonomies and generic skills	T&L methods	Assessment methods
CLO1	Recommend resolutions to the ethical issues of Artificial Intelligence that are raised by the development, deployment and use of AI	PLO2 (C2)	C5 S	Lecture Think pair share Case Study	F T
CLO2	Explain methodologies and key approaches to AI ethics with a particular emphasis on virtue ethics to support and enhance human well-being in a sustainable environment	PLO5 (C3C)	A4 SI	Peer Assessment Reflection	Pr Asg
CLO3	Practise the appropriate ethical risk assessment and management using relevant framework according to the relevant ethics and professional code of practice	PLO11 (C5)	A5 A	Brainstorming Case Based Learning	CADR

COURSE OVERVIEW



Course Synopsis



Course Learning Outcomes



Course Evaluation



Course Contents

SAIA1013 - RESPONSIBLE AND ETHIC (Section 01)

Lecturer:

- Dr Nurazean Maarop (016-9853093) (MJIIT 060601)

Venue:

- Lecture Room 1, Level 15, MR

Day:

- Slot 1: Monday (10:00am – 12:00pm) – 2 hours
- Slot 2: Wednesday (2:00pm – 4:00pm) – 2 hours

	Assessment	%
1	Assignment (5,5,10)	20
2	Presentation	10
3	Case Analysis and Discussion	25
4	Test	15
5	Final Exam	30
	Total	100

COURSE OVERVIEW



Course Synopsis



Course Learning Outcomes



Course Evaluation



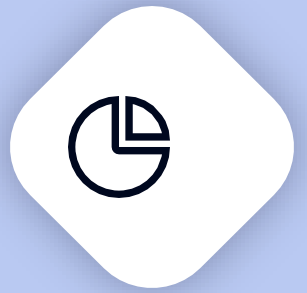
Course Contents

Week	**Assessment**
1	Introduction to Responsible AI and AI Ethics (lecture – synchronous, independent-study)*
2	Theoretical Concept of Ethics and their Application to AI (lecture (video) asynchronous, independent-study)
3	Values in AI Ethics (Islamic, national, others point of view) * (swap: Lecture on Wednesday)
4	Ethics Centric in AI Design*
5	Risk Assessment for AI Ethics*
6	Governance for Responsible AI*
7	Continuous Improvement & Presentation* (Seminar IEEE)

Teaching Plan

1. Mixture of Blended learning mode asynchronous (Individual study learn at your own pace) & synchronous (with lecturer)

2. Class engagement session



Independent Study (e.g.):

- What is AI?
- What is Responsible AI?



Individual /Group Output:

- Read the material assigned to you, find more related references and summaries your understanding



Critical thinking : (e.g.)

- Ethics in the Digital Era

MEETING 1 OVERVIEW

Introduction to Responsible AI and AI Ethics

(Understand the concept of Responsible AI and its significance in modern AI practices)

- Responsible AI & Principle
- Ethics in Digital Era
- The ART of AI (Accountability, Responsibility, Transparency)
- Three Level of AI Transparency (Independent-study)
- Bias in AI (Independent-study)
- Fairness and Transparency (Independent-study)
- Privacy and Surveillance (Independent-study)

(Some are in form of these activities: Lecture, Discussion, Case Studies, Short Quiz, Self Study)

SAIA 1013

By the end of the course, a student should be able to master some soft skills:

- Demonstrate knowledge of philosophical issues involved in ethics of artificial intelligence
- Demonstrate familiarity with relevant examples of AI systems
- Show ability to work in a small team or in pair
- Show ability to produce written work regularly to a deadline
- Acquire ability to express arguments clearly and concisely
- Gain skills in research, analysis and argumentation

Things to ponder from these 2 media content..

1. https://www.tiktok.com/@hitzdotmy/video/7522393736131775761?_t=ZS-8yP8xQ3tINs&_r=1

2. https://www.facebook.com/share/14EZd8SXztq/

a) What do you say after watching the above tiktok? Or read the facebook content?
b) What is the difference between responsible AI and AI Ethics, and how these two terms are connected based on the media contents?

Key Takeaway (case (b)):

✓ *AI Ethics guides why we should use AI responsibly: to save lives, avoid harm, and ensure fairness.*
✓ *Responsible AI is how we implement it: through transparent communication, human oversight, and equitable access.*

TODAY Lecture..



Understand the concept of Responsible AI and its importance in modern AI lifecycle



The Principles of Responsible AI

a) UNESCO

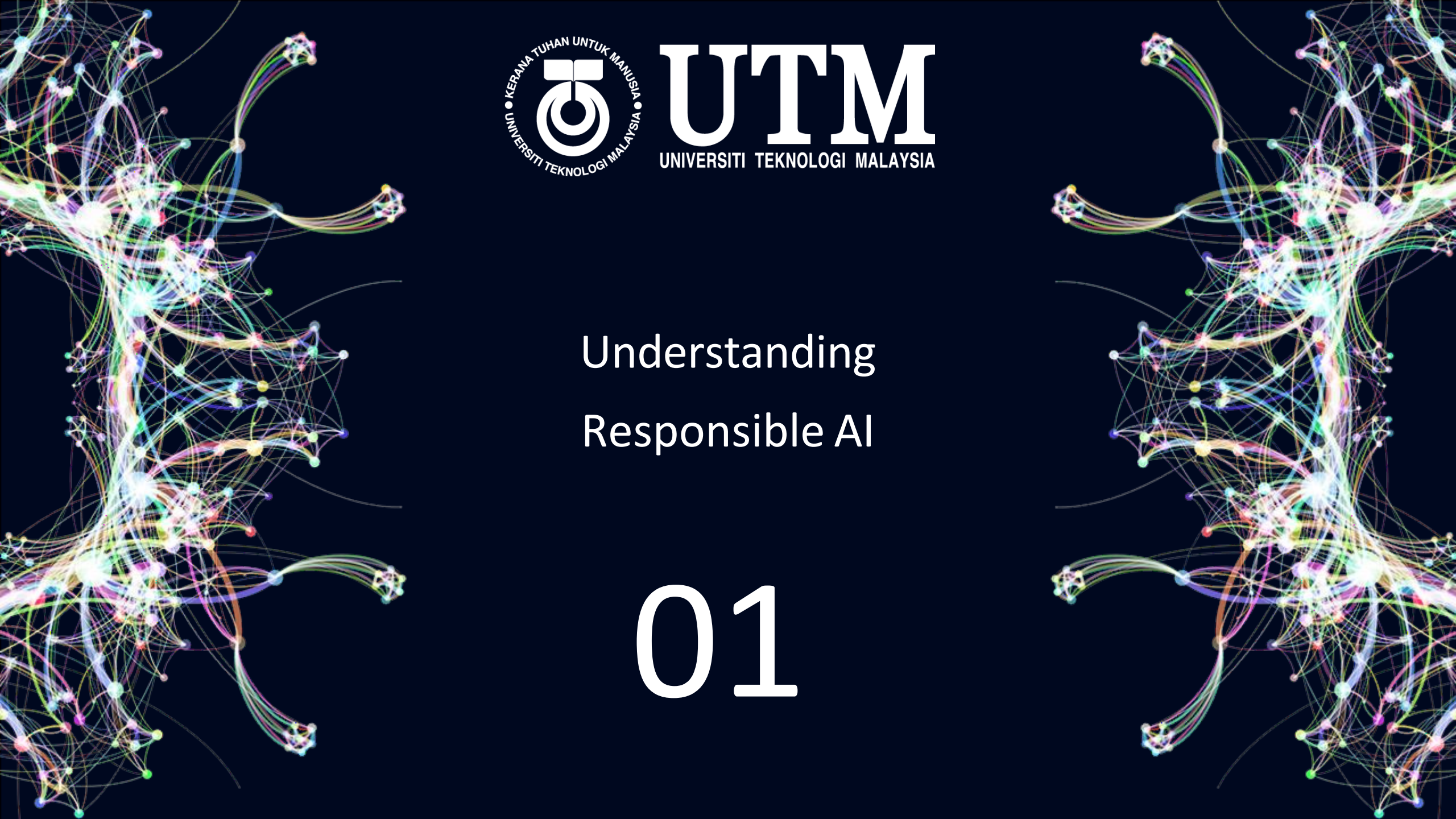
b) ASEAN

c) Malaysia



Ethics in Digital Era

But before that, what is “principle”?
Why it is important when it comes to subjective subject?



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Understanding Responsible AI

01

Introduction to Responsible AI and AI Ethics

- **Definition:** Responsible AI refers to the design, development, and deployment of AI systems in a way that is ethical, transparent, and aligned with human values.



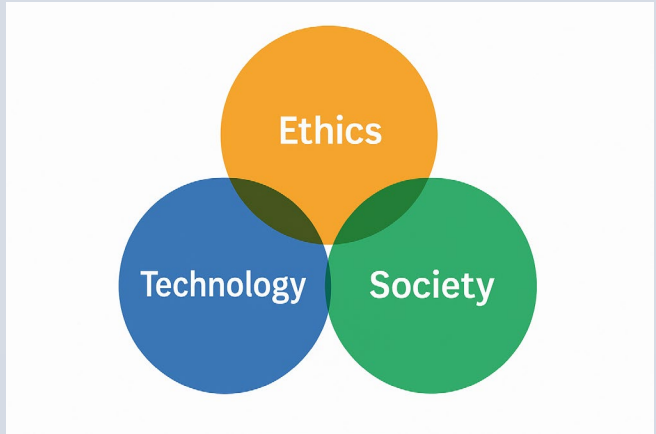
- **Some Key Characteristics (not limited to)**:
  - Human-centered: Designed to benefit people and society.
  - Ethically aligned: Respects moral values, norms, and rights.
  - Legally compliant: Adheres to laws, policies, and standards.
  - Transparent: Explainable and understandable by users.
  - Safe and secure: Minimizes risks and prevents harm.
  - Inclusive and Fair: Avoids bias and promotes equity across users.
  - Accountable: Developers and deployers take responsibility for outcomes.
  - Else??

Why Responsible AI Matters Now?

- Risks of irresponsible AI act (e.g., bias, misuse, lack of responsibility, accountability, transparency, explainability, privacy, data protection, autonomy and control)

- Rapid Digital Innovation: Technologies like Artificial Intelligence (AI), Internet of Things (IoT), and Big Data are reshaping every aspect of life.

- New Ethical Dilemmas such as:
  - Data Ownership: Who owns the data collected by AI?
  - Informed Consent: Do users understand how their data is used?
  - Algorithmic Bias: Are decisions fair and free from discrimination?
  - Accountability: Who is responsible when AI makes a harmful decision?

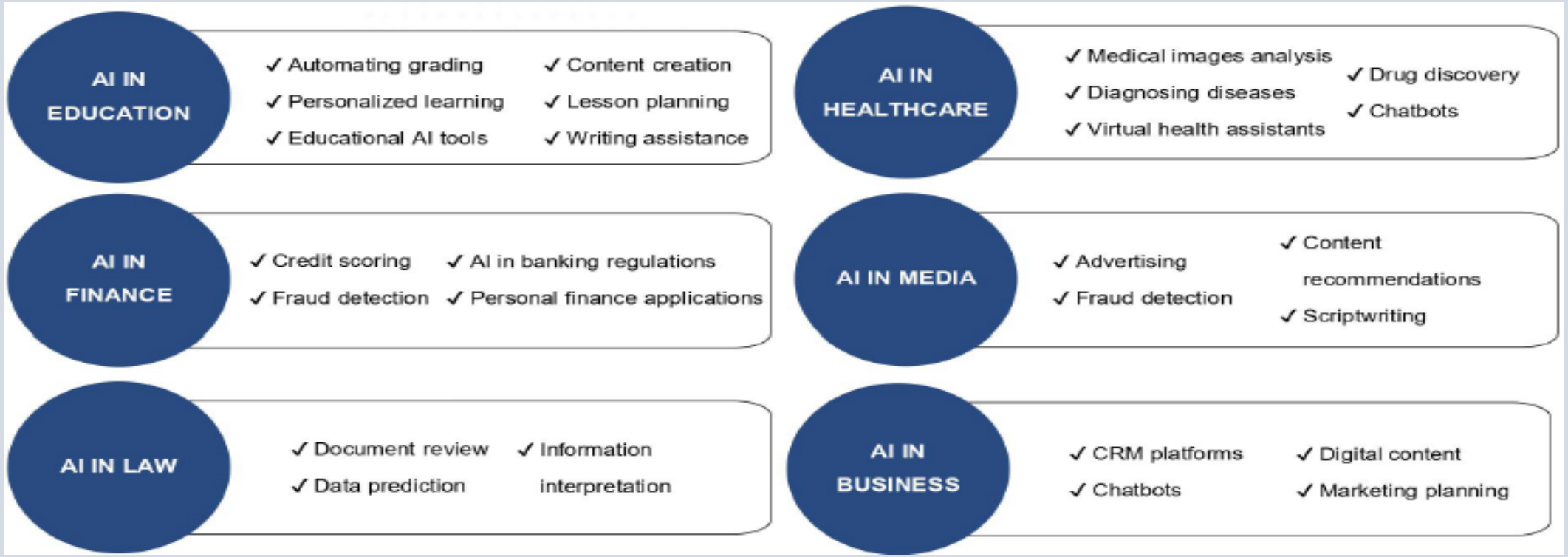


This Venn diagram shows how **Ethics**, **Technology**, and **Society** are deeply interconnected in the digital era.

- The **Technology** circle represents innovations like AI, big data, and IoT.
- The **Ethics** circle reminds us that these technologies must be guided by principles like fairness, accountability, and human rights.
- The **Society** circle highlights the people and communities affected by technology how it shapes lives, opportunities, and risks.
- AI that is technically robust, ethically sound, and socially responsible. If one circle is ignored say, technology is deployed without ethics we get outcomes like bias, privacy invasion, or loss of public trust.

Why it matters for the future?

- Links to sustainable development goal (SDG)
- Builds public trust (public confidence)
- Becoming the ‘norm’
- Choose one below, any real-world example of Irresponsible AI act you can think of? (5 minutes)



Source: CAIRO UTM

SDG 17: How SDG 17 Relates to Responsible AI?



The Sustainable Development Goals or Global Goals are a collection of 17 interlinked global goals designed to be a "blueprint to achieve a better and more sustainable future for all people and the world". The SDGs were set up in 2015 by the United Nations General Assembly and are intended to be achieved by the year 2030.

C6.0 ALIGNING INDUSTRIAL ECONOMIC CONTRIBUTIONS WITH RESPONSIBLE AI AND SDG & ESG PRINCIPLES

AI becomes a powerful tool that can drive sustainable development and contribute to the achievement of the Sustainable Development Goals (SDGs) while aligning with Environmental, Social, and Governance (ESG) principles. ESG compliance has become increasingly important in the face of increasing climate risks as it deals with a company's impact on the environment and society. ESG data includes any indicators that shed light into the sustainability context of an asset, facility, company or region, whether historic, current or expected. ESG data is collected under three primary umbrellas:

- Environmental data should capture environmental information such as annual carbon emissions and energy consumption, water usage, waste and pollution output, and more.
- Social data focuses on statistics related to workforce diversity, gender equity, human rights, and more.
- Governance data tracks company input regarding corruption, labor practices, the gender composition of the board of directors, etc..



Figure 13: Guidelines AI Governance and Ethics for each AI Socio Economic Drivers (2024-2025)

https://mastic.mosti.gov.my/publication/the-national-guidelines-on-ai-governance-ethics/

Common terms in AI world (recap)...



Artificial intelligence

Overall term describing a set of different kinds of techniques and technology to make machines and computers behave in some kind of intelligent fashion.

Key parts of AI is ability to perform tasks without supervision and to learn to improve performance



Machine learning

Machine learning is a set of algorithms which by themselves learn to make decisions or to structure data.

Supervised and unsupervised learning are based on data, while reinforcement learning is where the algorithm uses trial and error to learn to make sequences of decisions.



Generative AI

Models of algorithms that create new output such as text, video, photo, code, data, statistics from huge amounts of data that they are training of.

Major Artificial Intelligence Techniques
(that you may consider to use or develop with full responsible)

Technique	Description
Machine Learning	A branch of AI that uses data and algorithms to enable machines to learn like humans and improve over time. *(Source: IBM)*
Deep Learning	A subset of ML that uses multilayered neural networks to simulate complex decision-making like the human brain. *(Source: IBM)*
Computer Vision	Enables machines to analyze and understand images or videos, making decisions based on visual input using ML and neural networks. *(Source: IBM)*
Natural Language Processing (NLP)	A subfield of AI that enables machines to understand and generate human language using machine learning. *(Source: IBM)*

The ART of AI (Accountability, Responsibility, Transparency)

- A = Accountability → Who is liable?
- R = Responsibility → Who takes ownership?
- T = Transparency → Is the AI explainable?

Applying ART in Practice

- Case Study: AI in University Admissions
  - A university uses an AI system to screen and shortlist applicants for undergraduate programs based on grades, extracurriculars, and recommendation letters.
- How ART can prevent harm:

A – Accountability:

The admissions office assigns a human supervisor to oversee AI decisions. There's a clear escalation process if an applicant appeals.

R – Responsibility:

Developers and university staff take joint responsibility for outcomes. The institution regularly audits the model for fairness and bias.

T – Transparency:

Applicants are given reasons for rejection and key evaluation criteria. The AI system's decision-making logic is documented and explainable.

Ethics in the Digital Era



Needs to explore the ethical, social and legal aspects of AI systems

AI technologies shape our societies

AI is expected to bring significant and diverse benefits to society , from greater efficiency and productivity to tackling a number of difficult global problems

HOWEVER

These technologies have potential to produce undesirable impacts:

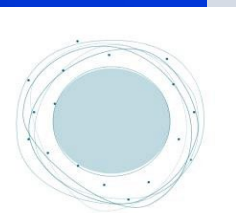
- Algorithms can enhance already existing biases
- Algorithms can discriminate
- Algorithms can threaten our security, manipulate us and have lethal consequences

What is Ethics...

→ answer questions like "what is good or bad", "what is right or what is wrong", or "what is justice, well-being or equality". Ethics is a set of beliefs about right and wrong behavior within a society.

→ As a discipline, ethics involves systematizing, defending, and recommending concepts of right and wrong conduct by using conceptual analysis, thought experiments, and argumentation.

→ The goal: To help people to make moral decisions



01

Meta-ethics

Meta-ethics studies the meaning of ethical concepts, the existence of ethical entities (ontology) and the possibility of ethical knowledge (epistemology).

02

Normative ethics

Normative ethics concerns the practical means of determining a moral (or ethically correct) course of action.

03

Applied ethics

Applied ethics concerns what a moral agent (defined as someone who can judge what is right and wrong and be held accountable) is obligated or permitted to do in a specific situation or a particular domain of action

Ethical frameworks are attempts to build consensus around values and norms that can be adopted by a community – whether that's a group of individuals, citizens, governments, businesses within the data sector or other stakeholders.

Evolution of Terminologies

- Computer ethics
- IT ethics
- Internet ethics
- ICT ethics
- Information ethics
- Cyber ethics
- Responsible AI & Ethics

Responsibility : Legal VS Moral



Legal

An actor is responsible for an event when a legal system is liable to penalise that actor for that event.



Moral

An actor is responsible for an act, if they can be blamed for the action. Moral and legal responsibility are different things



Action

Primarily responsible for their own actions

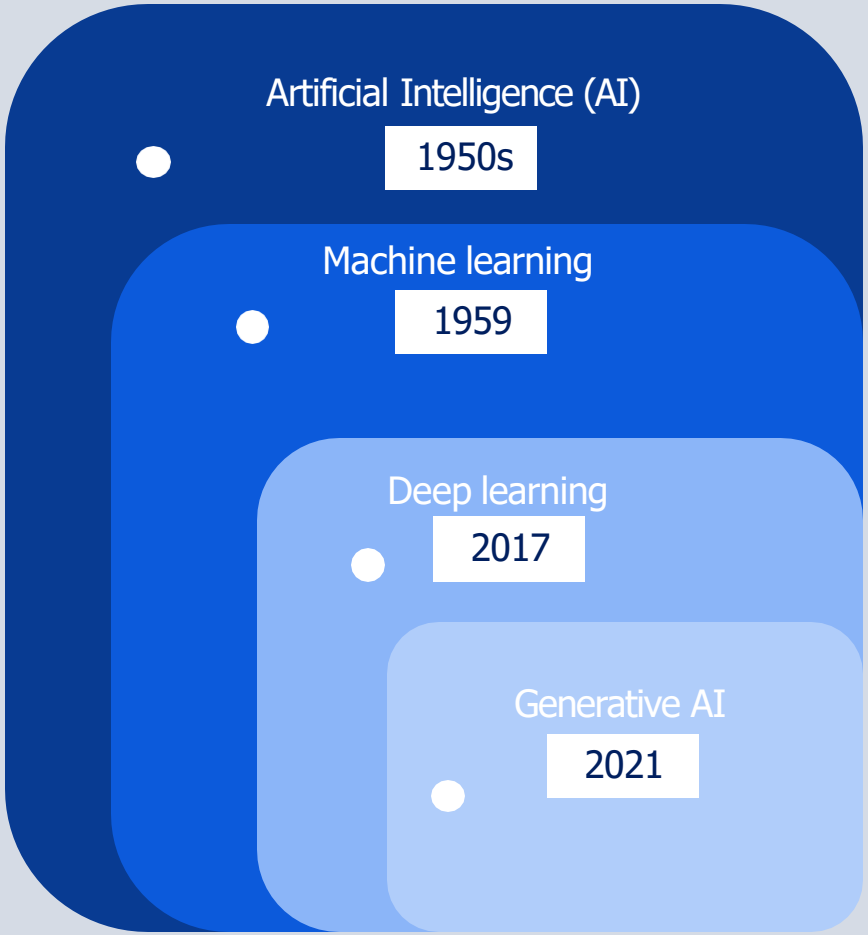


Omission

I am responsible for not taking action

Type	Explanation	Example in AI
Action	Responsible for what you actively do	Creating an algorithm that discriminates
Omission	Responsible for **not** acting when you should have	Failing to fix a known bug that causes harm

Related Evolution



Responsible AI



Systems designed to align with ethical principles, legal standards, and societal values

approach to developing and deploying AI systems in an ethical, legal, and trustworthy manner

How AI Works

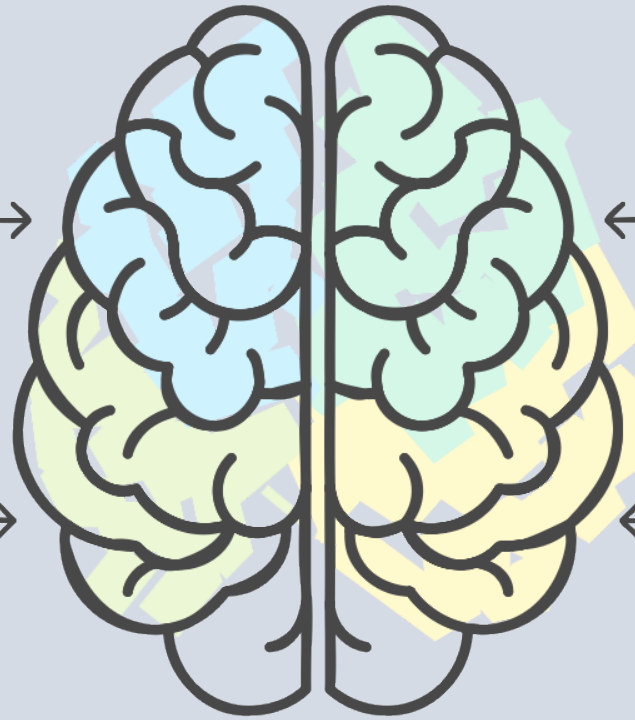
Characteristics of AI Systems

AI systems are modeled after the human brain's neural networks, particularly in deep learning.

Unlike traditional computing (sequential), AI processes multiple inputs simultaneously, enabling faster pattern detection.

Brain-inspired processing

Pattern recognition



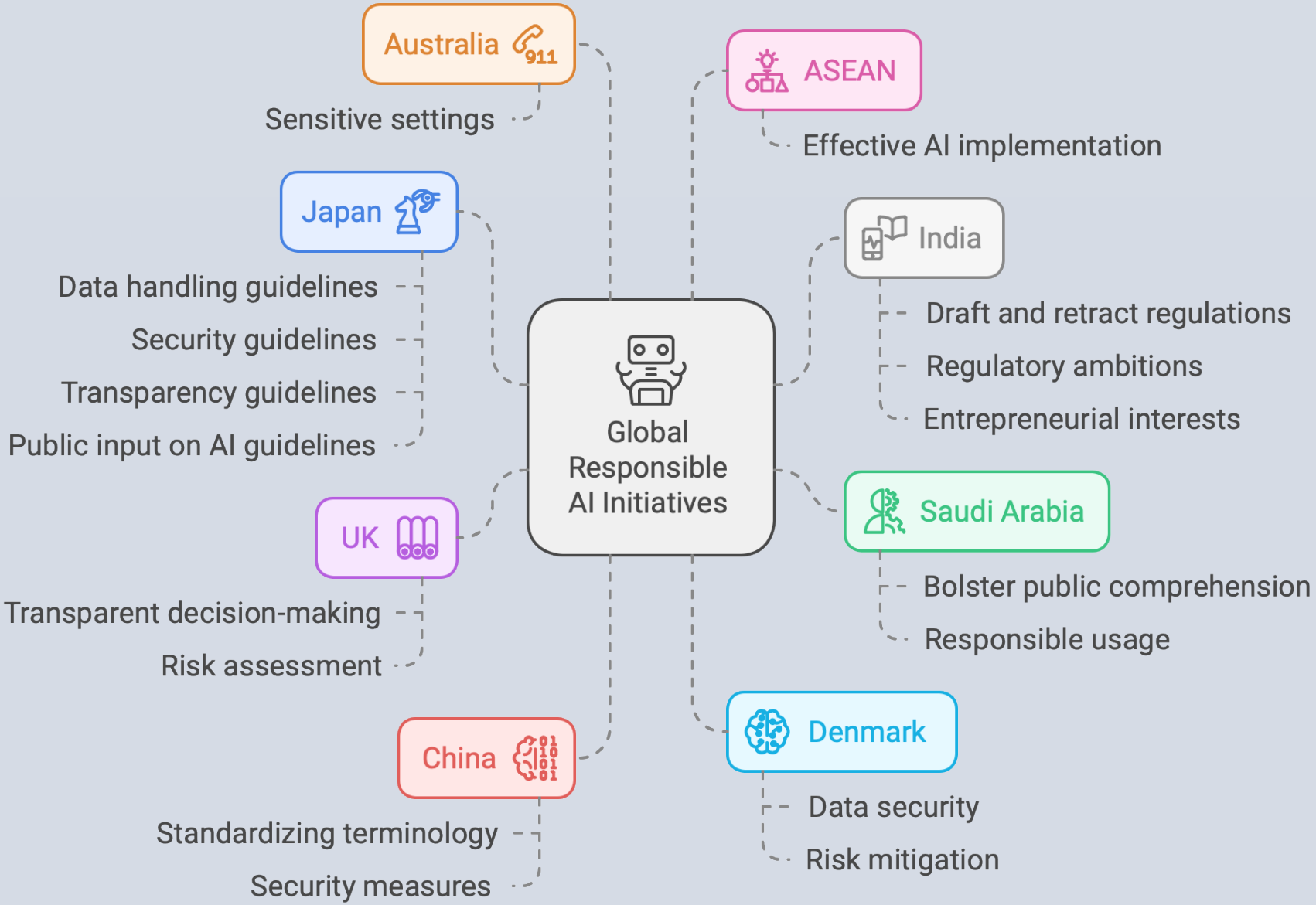
Adaptability

Application in machine/deep learning

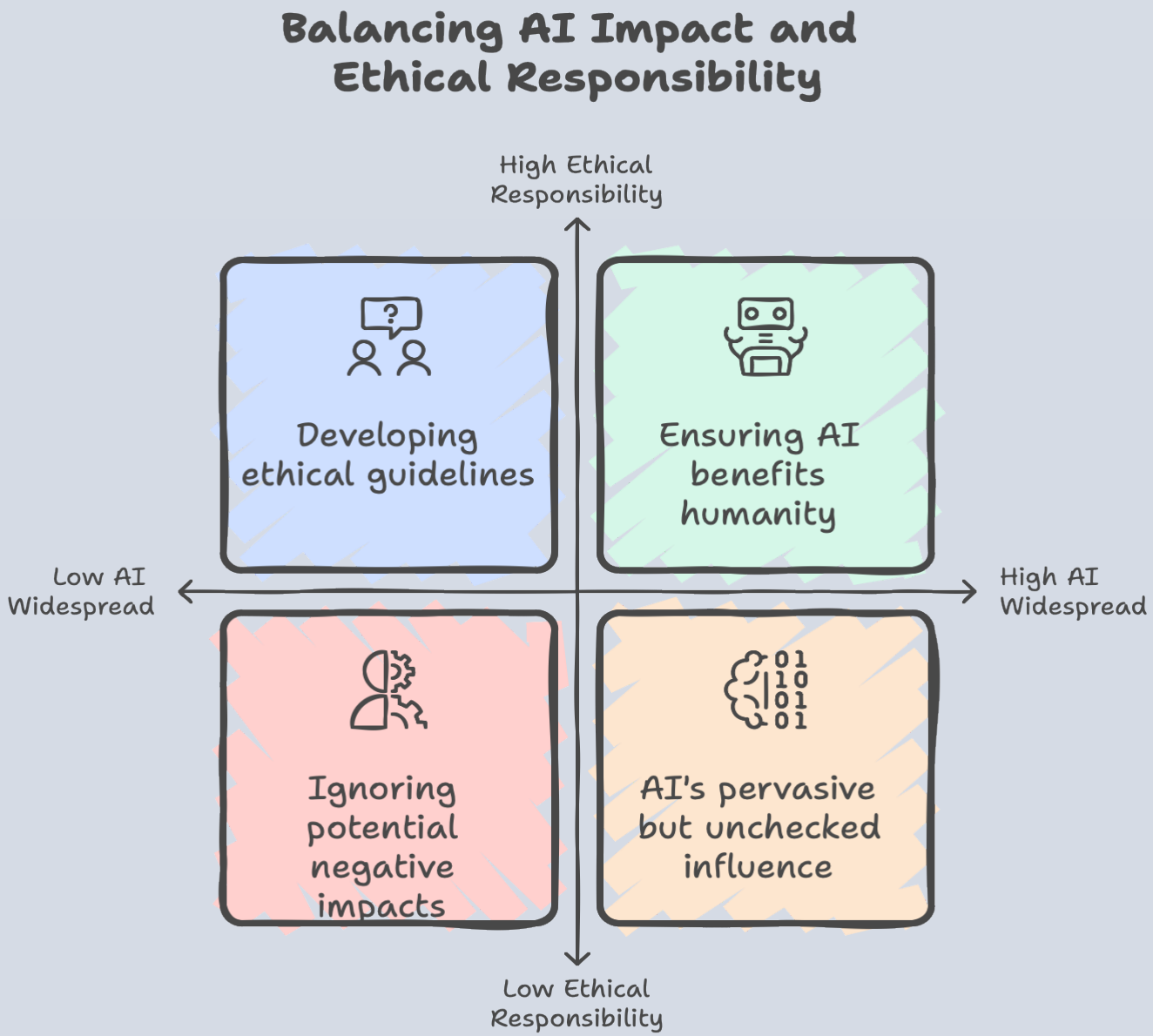
AI systems improve over time by learning from new data without explicit reprogramming Netflix's recommendation system adapts based on user watch history, refining suggestions over time.

Uses statistical models to recognize patterns (e.g., predicting stock prices) & uses deep neural networks for high-dimensional data.

Responsible AI Initiatives



Ethical Responsibility



Ideal Scenario (High Ethical Responsibility + High AI Widespread):
Developing ethical guidelines (e.g., fairness, transparency, accountability).
Ensuring AI benefits humanity (e.g., healthcare, education, sustainability).

Risk Scenario (Low Ethical Responsibility + High AI Widespread):
Ignoring negative impacts (e.g., bias, job displacement, privacy violations).
AI’s influence becomes unescapable but unchecked, leading to societal harm.



Principles of Responsible AI at UNESCO, ASEAN, and Malaysia Levels

02

UNESCO Recommendation on the Ethics of AI

2021



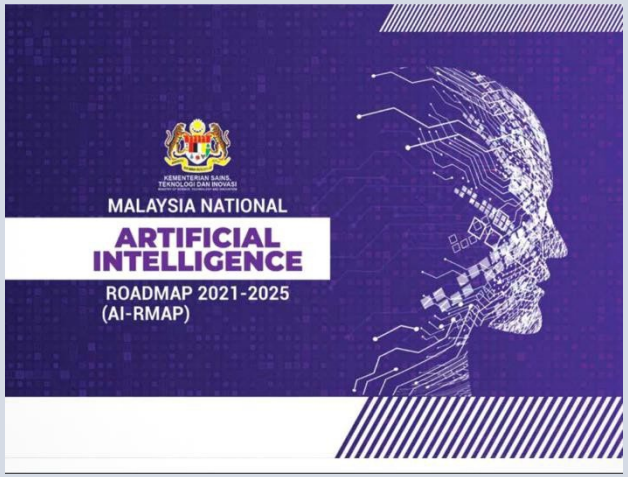
- The Recommendation on the Ethics of AI (UNESCO, 2021) which is the first-ever global standard on for AI Ethics was adopted by all 193 member states on 23 November 2021. The recommendation emphasises 10 AI Principles
- The protection of **human rights and dignity** is the cornerstone of the Recommendation, based on the advancement of fundamental principles such as transparency and fairness.
- **This recommendation** enables policymakers to translate the core values and principles into action with respect to data governance, environment and ecosystems, gender, education and research, and health and social well-being etc.
- UNESCO encouraged each of the country to develop AI governance for their country.

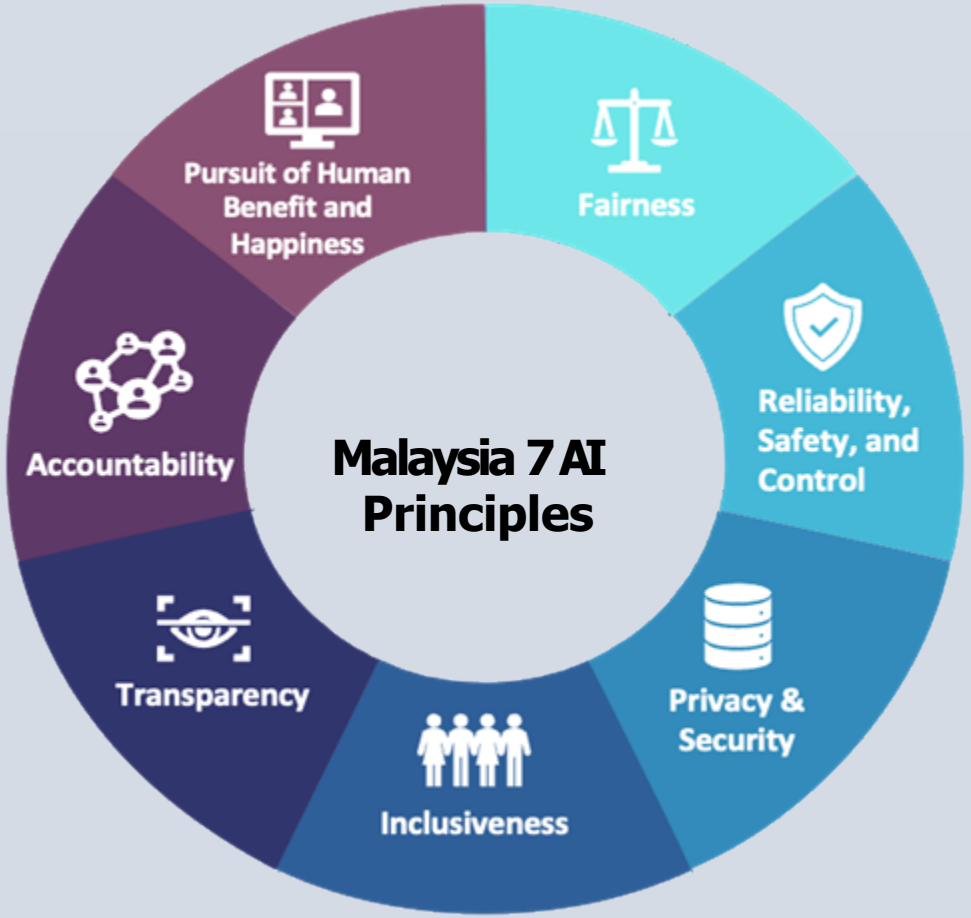
https://mastic.mosti.gov.my/publication/the-national-guidelines-on-ai-governance-ethics/

Malaysia Alignment with Global Calls : The AI Roadmap

2022

- To **support** the **UNESCO recommendation on AI ethics**, the Malaysian government introduced 7 AI Principles in August 2022 as explained in section 2.7.1
- The principles are included in the Malaysian National Artificial Intelligence Roadmap 2021-2025 (AI-RMAP)
- The Malaysian government has also taken the initiative to develop these Guidelines with the intention to **enhance AI technology development and adoption.**

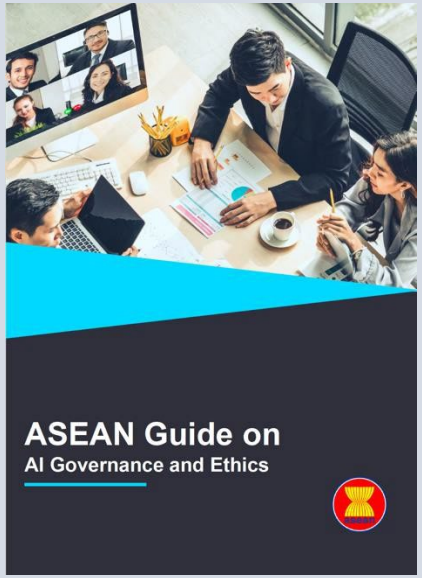




https://mastic.mosti.gov.my/publication/ artificial-intelligence-roadmap-2021-2025/

Recommendation on ASEAN AI Governance and Ethics Guidelines

2024



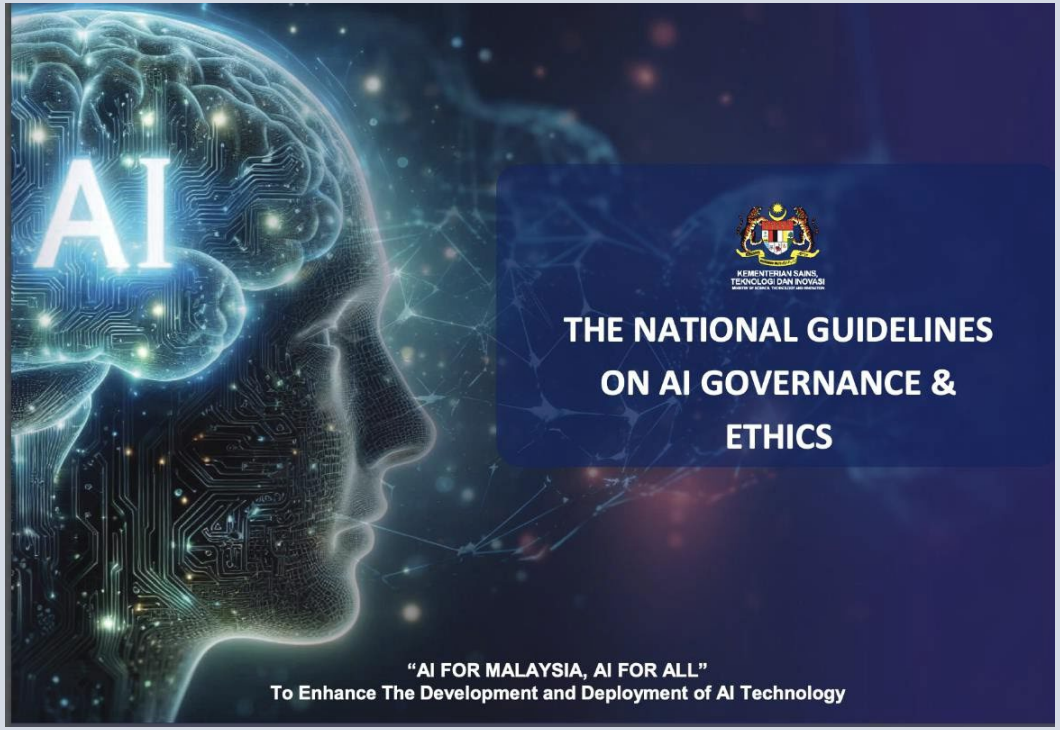
- At the regional level, ASEAN Member States have developed the ASEAN Guide on AI Governance and Ethics in February 2024 to empower organizations and governments in the region to use AI responsibly and increase users' confidence in AI.
- This document serves as a practical guide for organizations in the region that wish to develop and deploy AI technologies.



https://mastic.mosti.gov.my/publication/the-national-guidelines-on-ai-governance-ethics/

The National Guidelines AIGE

The National Guidelines AIGE- Sept 2024



https://mastic.mosti.gov.my/publication/the-national-guidelines-on-ai-governance-ethics/

2024

End Users

Government Agencies/ Policy Maker

Developers & Designer

Part A:

- Individuals or organizations who use AI products

Part B:

- Policy Makers in AI sub areas and features
- Government Agencies that used or potentially utilised AI as part of the process and optimisation
- Sector Players Industries/ Associations that regulate their own value chain

Part C:

- Contractors, Vendors and Consultants Custom and tailor made technology solutions that are developed using AI with the specifications pre determined by a Third Party (policy maker, etc)
- Developers and Solution Providers Innovation players that provide cutting edge solutions with the application of AI

The AIGE : Seven (7) AI Principles



1

FAIRNESS

The development of AI must be designed to avoid bias or discrimination against the target users. AI systems must not take a one-size-fits-all approach and needs to address the widest possible range of factors (e.g. age, gender, religion, and ethnicity). This enables equitable access and the active participation of all stakeholders.



3

PRIVACY AND SECURITY

AI systems should be safe and secure, and perform as intended, and resist being compromised by unauthorised parties. Developers should where necessary obtain consent from individuals before using or disclosing personal data for AI development and deployment. The systems must guarantee privacy and data protection throughout the entire life cycle of the AI system. Information and data collected from users must not be used in an unlawful or discriminatory way against them. Developers also should incorporate security-by-design and privacy-by-design principles and refer to international standards related to information security and privacy when implementing the AI system.



2

RELIABILITY, SAFETY AND CONTROL

AI systems or solutions must be robustly tested to be reliable, safe and fail-safe by default so that users can trust and depend on the AI solution. To ensure proper access, control and protection of critical situations, AI systems should work reliably, consistently and operate in a real world under normal circumstances and unexpected conditions. In order to prevent or mitigate negative outcomes, AI systems must be able to response quickly according to the intended purposes. Developers and end-users are encouraged to carry out relevant testing and certification as well as risk and impact assessments to prevent any potential harm and mitigate risks. Autonomous systems must have safeguards to ensure ultimate controllability by humans, particularly when it comes to the applications of AI in high-risk areas such as autonomous vehicles, military applications, and when human life is at stake.

https://mastic.mosti.gov.my/publication/the-national-guidelines-on-ai-governance-ethics/

Seven (7) AI Principles



4 INCLUSIVENESS

AI must be inclusive for all Quadruple Helix stakeholders in order to avoid unequal access to AI as in the case of social clefts between "Digital Haves" and "Digital Have-Nots"

AI systems should benefit everyone and address national needs and experience inclusively in compliance with the Federal Constitution and National Principles. This principle should be adhered in 3 ways: using inclusive development techniques; strategically developing tools which are sensitive to the specific needs of vulnerable groups in society; and proactively ensuring diversity amongst AI developers and decision makers.



5 TRANSPARENCY

AI algorithms should be transparent to ensure that any capabilities can be explained, both technical processes of an AI systems and related human decisions. This will allow stakeholders to evaluate the risks of AI and address issues that may arise.

Transparency principles are mainly applicable to contexts where AI is being used as part of decision-making process. There are 5 elements that need to be considered: full disclosure about the facts that an AI system is being used in decision making; the intended purpose of the AI systems; the training data (describing the data that was used in training, what historical and social biases exist in the data; what procedures were used to verify data quality); maintenance and assessment of AI systems; and the ability to challenge the decisions made by the AI systems.

https://mastic.mosti.gov.my/publication/the-national-guidelines-on-ai-governance-ethics/

Seven (7) AI Principles



6 ACCOUNTABILITY

The developers, owners of AI models, and AI actors should be accountable for the success or failure of the AI solutions. They should take responsibility for ensuring the proper functioning of AI systems in compliance with AI Acts, governances, and ethical principles.

While designing AI systems, 4 elements (system purpose, technology capability, quality and reliability, and sensitive users) need to be considered to avoid consequential harm.



7 THE PURSUIT OF HUMAN BENEFIT AND HAPPINESS

AI systems should respect human-centered values, pursue human benefit for human society, enhance the quality of life, and increase human happiness.

AI systems should not be used for malicious purposes in making decisions. Understanding the way users interact with AI systems and how it is perceived if there are any negative outcomes from the adoption of AI can improve happiness and human well-being.

Human oversight helps to ensure that AI systems do not undermine human autonomy through governance mechanism such as human-in-the-loop (HITL), human-on-the-loop (HOTL) and human-in-command (HIC). HITL refers to the capability for human intervention in every decision cycle of the AI systems, while HOTL refers to the capability for human intervention during the design cycle of the system and monitoring the system's operation. HIC refers to the capability to oversee the overall activity of the AI systems and the ability to decide when and how to use the AI system.

https://mastic.mosti.gov.my/publication/the-national-guidelines-on-ai-governance-ethics/

Summary of 7 Principles

Alignment with AI Ethics

	7 AI Principles/ Technology	AI Technology	Software Solution
1	Fairness	No biasedness in data collections, datasets, and algorithms are used to develop AI models and Use-Cases	Ensuring that the software processes and solutions are accessible by all users equitably
2	Reliability, Safety and Control	Robustness AI's pillars for AI Models (e.g Volume, Velocity, Variety, Variability, Veracity, Value & Visualization)	Ensuring that the solution performs its intended function under the stated conditions for a specified period of time without problems
3	Privacy & Security	Data Collections remain encrypted and confidential and in compliance with the PDPA Act (if personal records) or BNM regulations for others	Protecting the confidentiality, integrity, and availability of information in compliance with PDPA act including complete with appropriate Disclaimer regulated by law (BNM)
4	Inclusiveness	Customized and automated-ML, NLA, NLP, with solutions tailor made for user needs.	Creating software that is accessible from different platform such as web and mobile
5	Transparency	Disclosing limitations (e.g. ML, NLA, NLP) and potential problems with the AI models	Developers should be open about the capabilities and limitations of the software solution
6	Accountability	Adopting to best practice on designing, developing and deploying AI models with 95% accuracy	Adopting to best practice and perform regular source code vulnerability scanning
7	Pursuit of Human Benefit and Happiness	Regularly review, evaluate and enhance to ensure AI models align with human values and ethical considerations.	Ensuring the solution is accessible, user-friendly, and makes a positive contribution

https://mastic.mosti.gov.my/publication/the-national-guidelines-on-ai-governance-ethics/

End Users

Do

1 Understand Your Needs
Clearly identify problems you want to solve with AI and understand how AI can specifically address those needs.

2 Explore
Begin exploring the AI technology to know and to be competitive

3 Protect Personal Data
AI's performance is heavily depending on the availability of data from various sources. Therefore protecting individual personal data is crucial

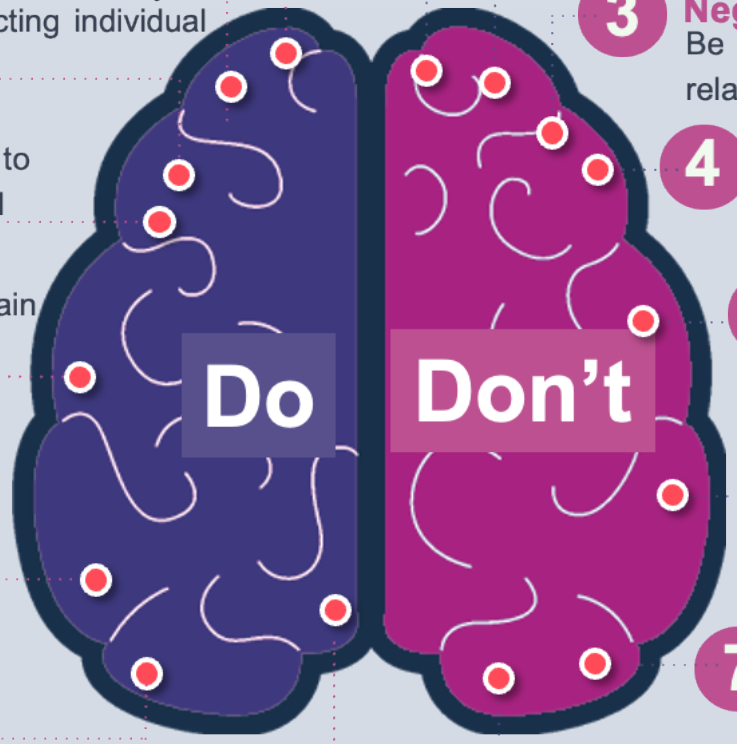
4 Adopting Ethical Guidelines
Adopting ethical guidelines and checks to ensure that the AI system is fair and unbiased

5 Upgrade AI Skills
Educate and train ourselves with AI to remain competitive

6 Stay Updated
AI is a rapidly evolving field. Regularly update your knowledge and skills to keep up with new developments.

7 Accepting the benefits of AI
To recognize and accept AI outstanding

8 Invest in AI
Start invest in AI and carry out assessment for future benefits and impact assessment for future investment



Don't

1 Underestimate the Complexity
Implementing AI is not just about technology; it involves change lifestyle and cultural shift.

2 Ignore the Human Element
AI should augment human abilities, not replace them.
Focus on how AI can assist and empower your daily life.

3 Neglect Data Privacy
Be mindful of privacy laws and ethical considerations related to data collection and usage.

4 Overlook Safety and Security
Comply to relevant regulation to safeguard our personal information and data

5 Chase Hype
Don't adopt AI just because it's a buzzword. Ensure it genuinely adds value to your wellbeing and happiness

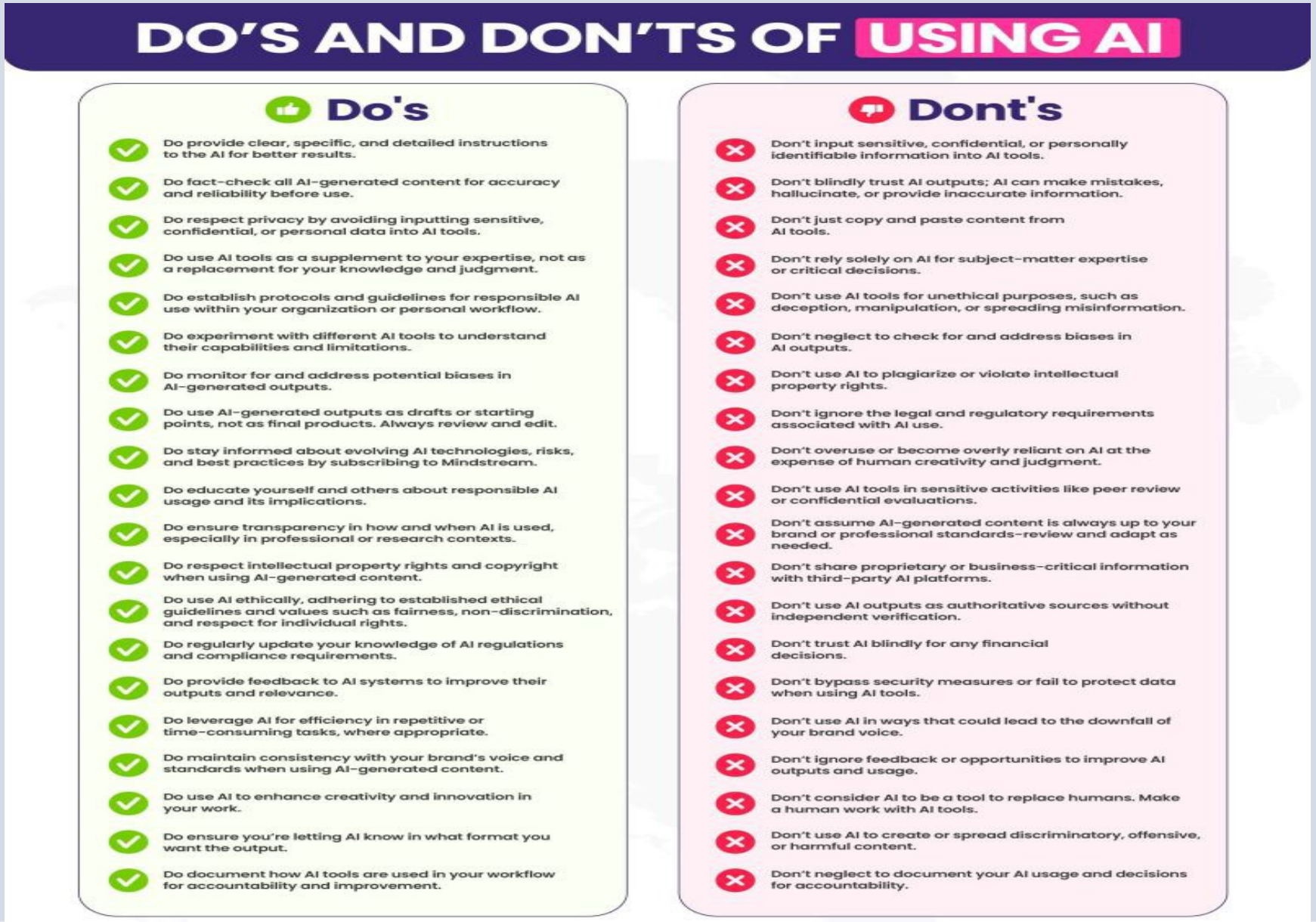
6 Overdependent on AI
Don't implement AI without understanding of how it fits into your lifestyle and avoid rely solely on AI

7 Ignore on Reliable Source
Don't adhered and trust unreliable data and source

8 Disregard Regulations
Stay compliant with all relevant laws and regulations affecting AI in your industry and region.

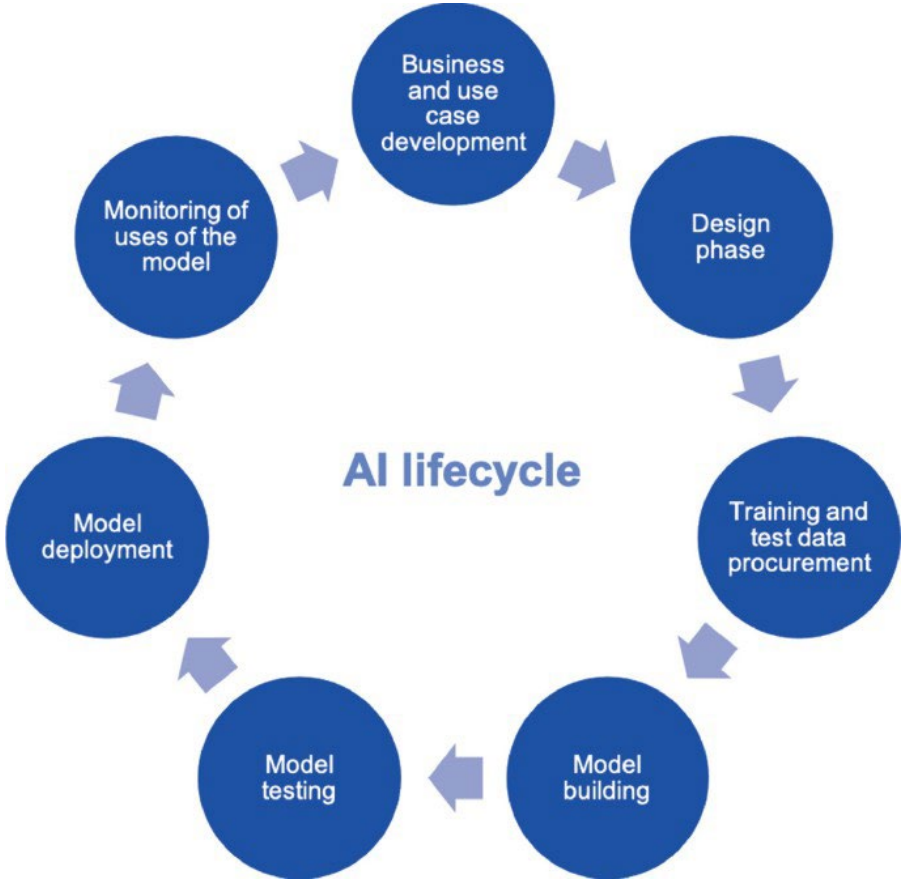
https://mastic.mosti.gov.my/publication/the-national-guidelines-on-ai-governance-ethics/

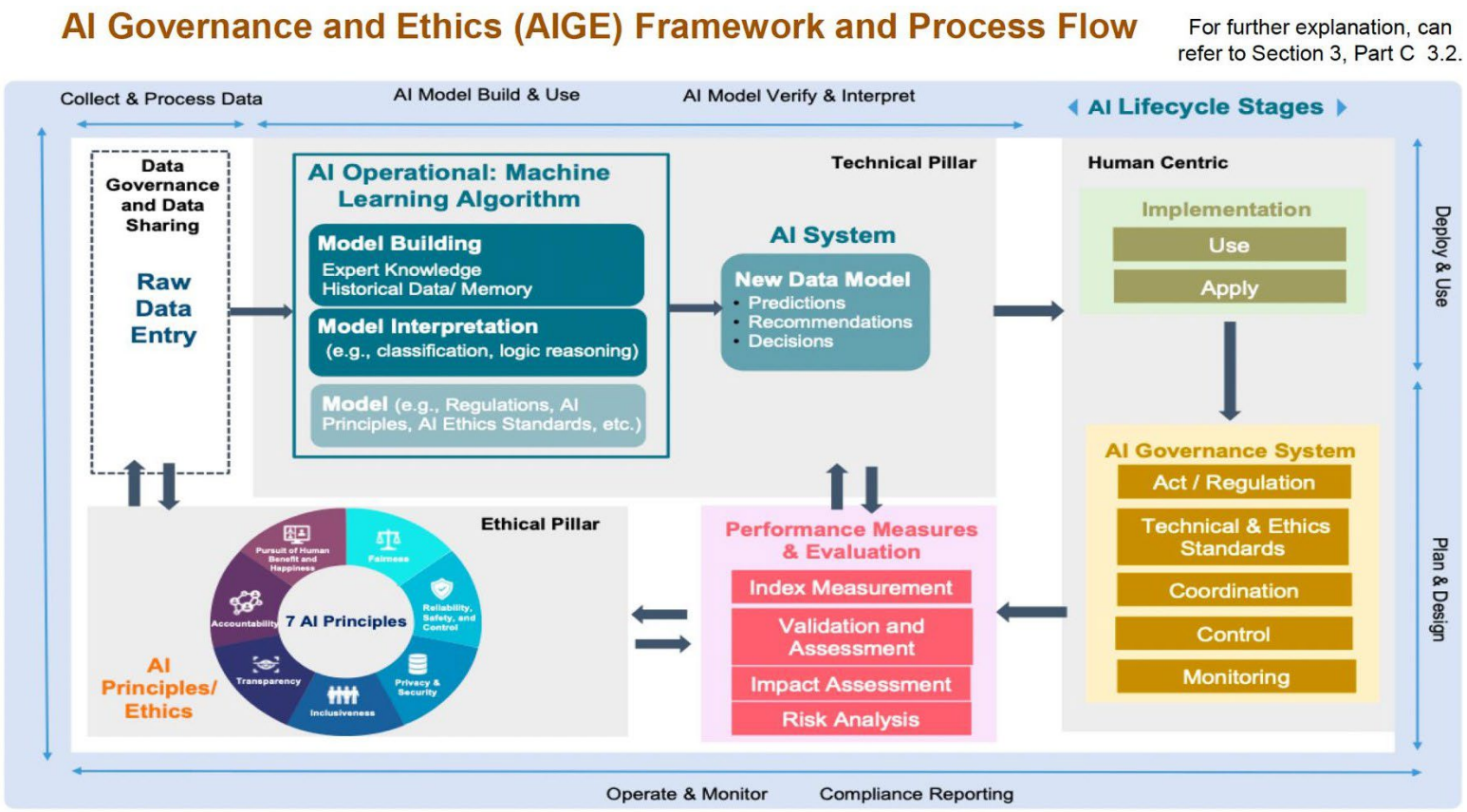
Do and Don’t


DO’S AND DON’TS OF USING AI
Do's
Do provide clear, specific, and detailed instructions to the AI for better results.
Do fact-check all AI-generated content for accuracy and reliability before use.
Do respect privacy by avoiding inputting sensitive, confidential, or personal data into AI tools.
Do use AI tools as a supplement to your expertise, not as a replacement for your knowledge and judgment.
Do establish protocols and guidelines for responsible AI use within your organization or personal workflow.
Do experiment with different AI tools to understand their capabilities and limitations.
Do monitor for and address potential biases in AI-generated outputs.
Do use AI-generated outputs as drafts or starting points, not as final products. Always review and edit.
Do stay informed about evolving AI technologies, risks, and best practices by subscribing to Mindstream.
Do educate yourself and others about responsible AI usage and its implications.
Do ensure transparency in how and when AI is used, especially in professional or research contexts.
Do respect intellectual property rights and copyright when using AI-generated content.
Do use AI ethically, adhering to established ethical guidelines and values such as fairness, non-discrimination, and respect for individual rights.
Do regularly update your knowledge of AI regulations and compliance requirements.
Do provide feedback to AI systems to improve their outputs and relevance.
Do leverage AI for efficiency in repetitive or time-consuming tasks, where appropriate.
Do maintain consistency with your brand's voice and standards when using AI-generated content.
Do use AI to enhance creativity and innovation in your work.
Do ensure you're letting AI know in what format you want the output.
Do document how AI tools are used in your workflow for accountability and improvement.
Dont's
Don't input sensitive, confidential, or personally identifiable information into AI tools.
Don't blindly trust AI outputs; AI can make mistakes, hallucinate, or provide inaccurate information.
Don't just copy and paste content from AI tools.
Don't rely solely on AI for subject-matter expertise or critical decisions.
Don't use AI tools for unethical purposes, such as deception, manipulation, or spreading misinformation.
Don't neglect to check for and address biases in AI outputs.
Don't use AI to plagiarize or violate intellectual property rights.
Don't ignore the legal and regulatory requirements associated with AI use.
Don't overuse or become overly reliant on AI at the expense of human creativity and judgment.
Don't use AI tools in sensitive activities like peer review or confidential evaluations.
Don't assume AI-generated content is always up to your brand or professional standards-review and adapt as needed.
Don't share proprietary or business-critical information with third-party AI platforms.
Don't use AI outputs as authoritative sources without independent verification.
Don't trust AI blindly for any financial decisions.
Don't bypass security measures or fail to protect data when using AI tools.
Don't use AI in ways that could lead to the downfall of your brand voice.
Don't ignore feedback or opportunities to improve AI outputs and usage.
Don't consider AI to be a tool to replace humans. Make a human work with AI tools.
Don't use AI to create or spread discriminatory, offensive, or harmful content.
Don't neglect to document your AI usage and decisions for accountability.

AI Governance & Ethics and AI Lifecycle

Who do you think bears the most responsibility in AI development?

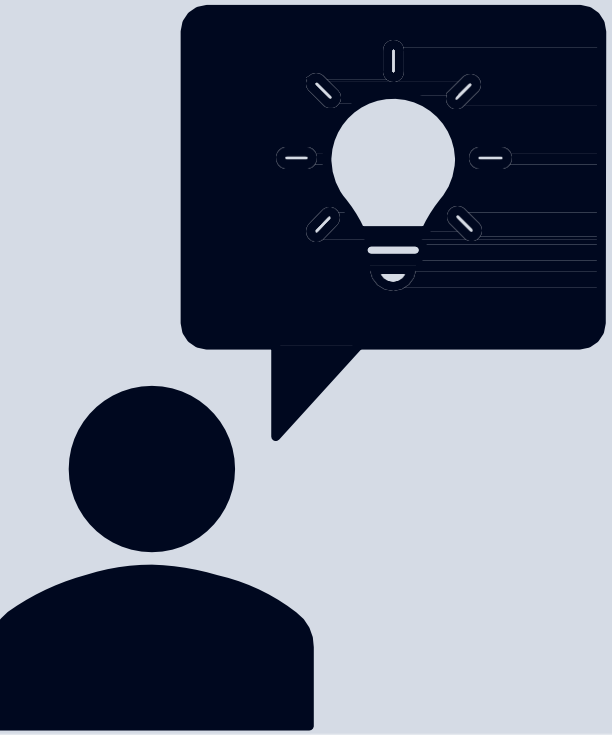




Jianlong Zhou, Fang Chen, Humanity Driven AI, 2022 ISBN : 978-3-030-72187-9
https://link.springer.com/chapter/10.1007/978-3-030-72188-6_1

https://mastic.mosti.gov.my/publication/the-national-guidelines-on-ai-governance-ethics/

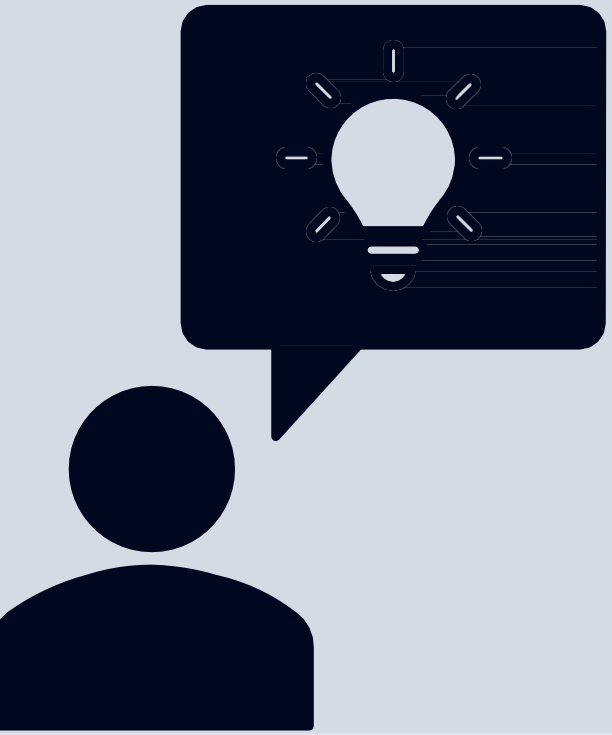
Stakeholder Responsibility in AI



- Shared responsibility across the AI lifecycle
- Roles of developers, users, and policymakers
- Importance of interdisciplinary collaboration

Your Responsibility in Ethical AI

Your Role in Ethical AI



- Opportunities to influence ethical AI development and use
- Importance of individual awareness and action
- Ongoing learning and engagement with AI ethics

Path Forward



EMERGING ETHICAL CHALLENGES IN AI



POTENTIAL FOR AI TO ENHANCE ACCOUNTABILITY AND TRANSPARENCY



THE EVOLVING LANDSCAPE OF AI RESPONSIBILITY



IMPORTANCE OF ONGOING ETHICAL DISCUSSIONS



ADVANCEMENTS IN EXPLAINABLE AI (XAI) AND PRIVACY-PRESERVING TECHNIQUES



GROWING EMPHASIS ON ETHICAL AI IN EDUCATION AND INDUSTRY

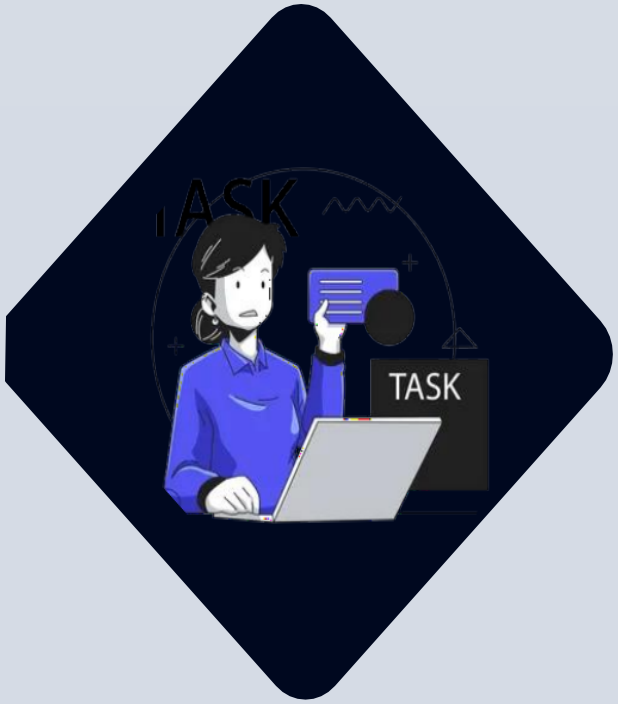


03

Critical Thinking:

Ethics in the Digital Era

Assignment Title: *Think-Pair-Share: Responsible AI Flash Cards* (5%)



Produce Flash Card teaching material

1. Work in pair
2. Choose 1 principles (can be from UNESCO, ASEAN or AIGE Guidelines)
3. Discuss and come out with minimum these 3 content:
   a) Brief introduction about the principle and the usage
   b) Give 1 case study that can explain about the ethics principle in digital era by giving i)before AI adoption and ii) how AI works & risk when issue arise
   c) Produce scenario with flash card quiz (there is question, there is answer)
4. Submit to e-learning platform. Attach minimum 5 slides (follow example)
5. If we have time, presentation will be picked randomly among pair group, or bonus will be given to volunteers.

Brief Explanation

ACCOUNTABILITY

In AI ethics, there are three different senses or dimensions of **accountability** based on different means of action including:

The question of determining the responsibility

→ Which individuals (or groups) are accountable for the impact of algorithms or AI?

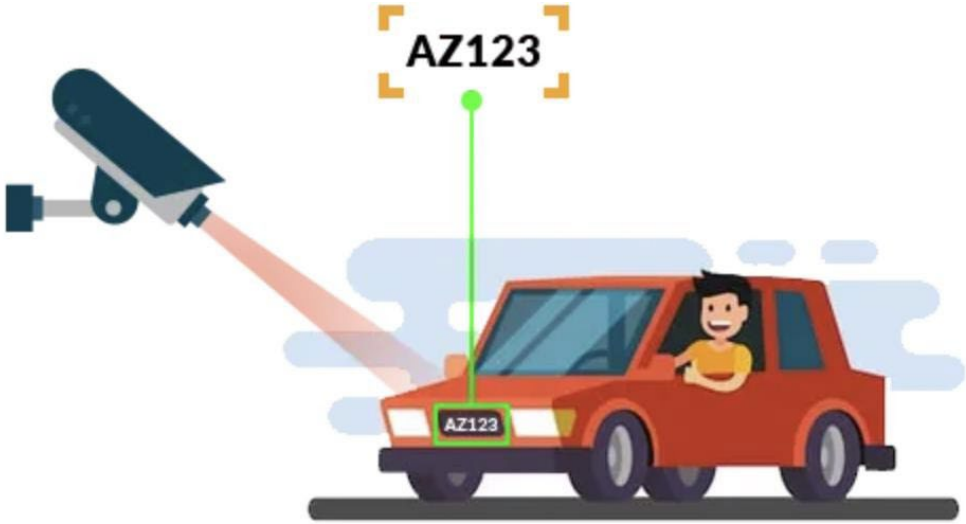
→ Who is responsible for what effect within the overall socio-technical system?

A feature of the societal system that develops, produces, and uses AI

A feature of the AI system itself

ACCOUNTABILITY

Case Study: Algorithms and accountability
Better service efficiency, reliability and consistency



Source: https://www.innominds.com/connected-devices-and-iot/numix-video-analytics-solutions/anpr

ANPR (Automatic number-plate recognition) based parking management system that charges you based on the duration of car is parked in the lot:

1. Camera takes a picture of the car.
2. Object detection model detect the license plate area of the image.
3. The license plate area of the image is cropped and sent to an OCR (Optical Character Recognition) algorithm that converts images into text.
4. The text is processed and matched against various templates to understand the license plate number.

ACCOUNTABILITY

Case Study: Algorithms and accountability
Better service efficiency, reliability and consistency

Entrance to mall process would take about a minute.
The line into the mall would interrupt traffic on the main road on holidays and weekends.





Scanning systems may malfunction, or suffer from bugs.
Issued invalid fee

Who should take the responsibility and on what grounds?

ACCOUNTABILITY

Organizations designing and deploying algorithms can be taken to be morally responsible through governance structures

The algorithms are causal factors that underlie the decisions; human that makes the final decision and also takes responsibility



ACCOUNTABILITY

Quiz: Deepfakes



Scenario

Imagine you're part of a popular university social media club, CampusConnect, which manages a platform where students share videos and updates about campus life.

Recently, a deepfake video of the university's president, Dr. Lee, was uploaded to the platform.

The video falsely showed Dr. Lee announcing a surprise tuition hike, causing widespread panic and protests among students.

ACCOUNTABILITY

Flash Cards

"CampusConnect's Tech Team decided not to rush the deployment of the deepfake detection tool, waiting for further testing."



Omission

Action

Level of Responsibility:
High: have potentially prevented the spread of the deepfake.

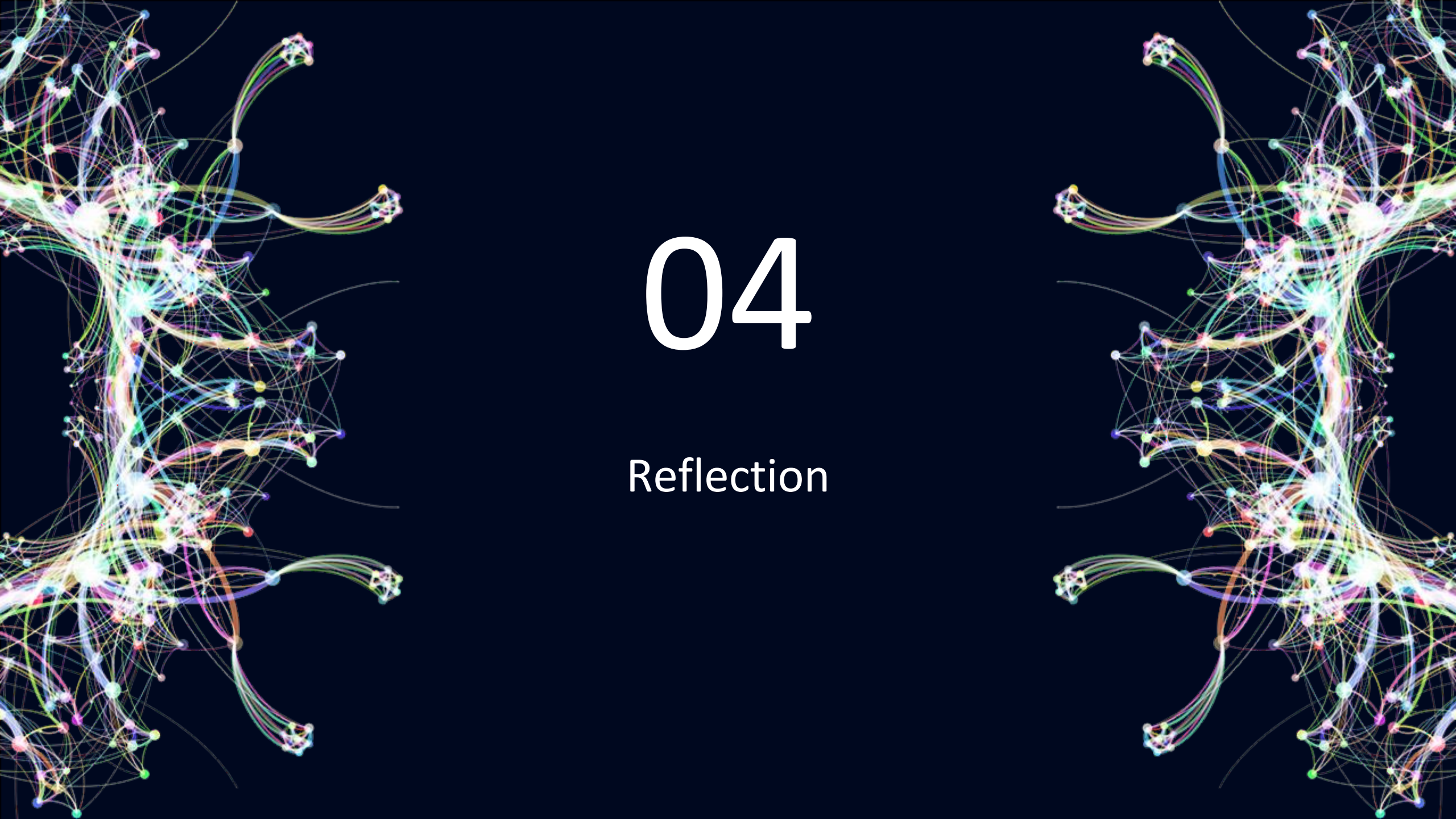
"CampusConnect's leadership chose to prioritize user freedom over implementing stricter content moderation policies."

Omission



Action

Level of Responsibility:
High: decision influenced the platform's ability to mitigate harm from deepfakes.



04

Reflection



How to ensure ethical AI development?

Ethics in AI revolves around fundamental design decisions that shape how these systems interact with and impact society

Responsible AI

Approach to developing and deploying AI systems in a safe, trustworthy, and ethical manner. It involves understanding responsibility for outcomes, ensuring transparency for end-users, and integrating ethical considerations throughout the AI system's lifecycle.

AI Ethics Principles

Guidelines intended to ensure responsible AI development and use. These principles are often abstract and domain-agnostic, requiring adaptation to specific contexts.

AI Ecosystems

Involve a complex interplay of various actors, including developers, users, policymakers, and the broader society. Understanding the ethical implications of AI requires considering the entire ecosystem and its dynamics.

Ethical Decision-Making

Explores methods for AI systems to make ethical decisions. Is AI systems can or should be capable of making ethical decisions? Different approaches to implementing ethical reasoning in AI, including top-down (rule-based), bottom-up (learning from data), and hybrid approaches.


KERANA TUHAN UNTUK MANUSIA
UNIVERSITI TEKNOLOGI MALAYSIA
UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Thank you for
your attention

Core Ethical Consideration by CAIRO UTM (Self-Study - eLearning)

Instruction

Read the material given in eLearning (self-study). The material is very brief around the 5 ethical consideration/implication aspects about AI, as shown in the below figure. Therefore, you need to find more information about these 5 ethical consideration /implications from any legit textbook or research material and the aim of the self-study are you should be able to:

1. Define the 5 terms as shown in the figure.
2. What are the implication of each aspect having on the education (be it on either: teacher, lecturer and student)
3. Write only 3 significant points from your self study using the e-learning forum (When contributing to the forum, avoid repeating points already mentioned by earlier participants. Instead, build on the discussion by adding new insights, perspectives, examples, or related information. Ensure your comment is concise, relevant to the topic, and offers additional value to the conversation) (e-learning reminder: You are required to manually mark the activity as 'done')

